

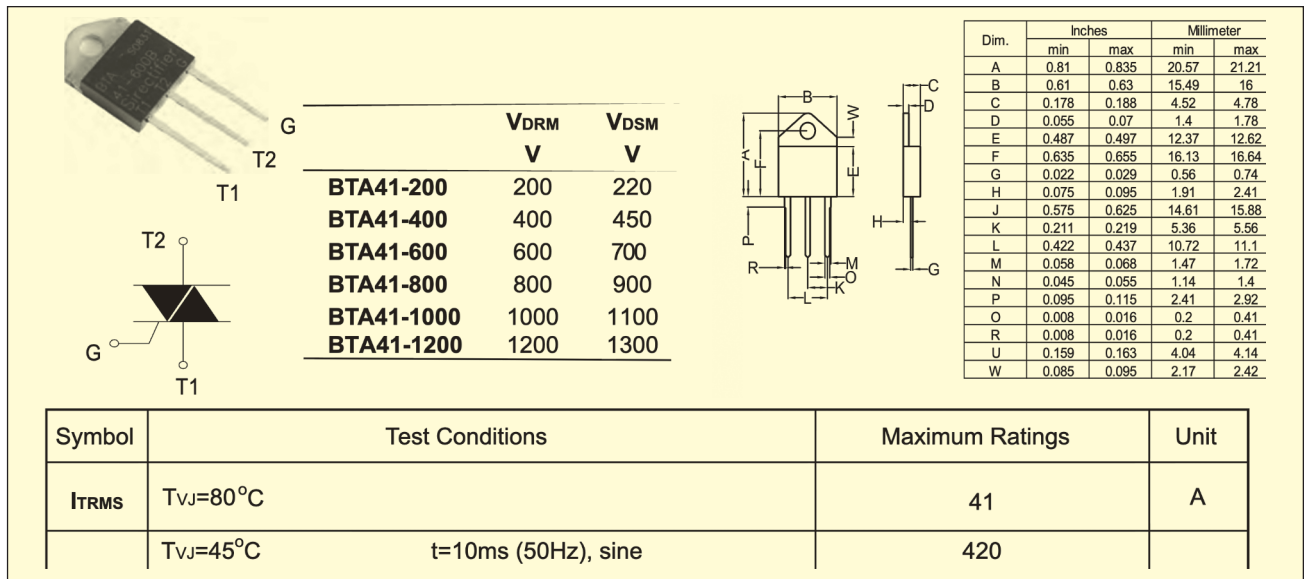


Fig. 3: Source: BTA41 IC datasheet

BILL OF MATERIALS					
Name	Designator	Footprint	Quantity	Manufacturer Part	Manufacturer
100n-400V polyester capacitor	C1	CAP-TH_L9.5-W4.0-P7.50-D0.6	1	82CL0114	KNSCHA
KF78C-13.0-2P	C2	CONN-TH_KF78C-13.0-2P	1	KF78C-13.0-2P	KEFA
100nF	C3	C1206	1	CC1206KKX7RYBB104	YAGEO
DIAC	D1	DIAC	1		
2-pin terminal connector 40A 800V	H1	CONN-TH_2P-P5.00_MX301- 5.0-02P-GN01-CU-Y-A	1	MX301-5.0-02P-GN01- Cu-Y-A	MAX
MOC3041	IC1	MOC3041	1	MOC3041M	FAIRCHILD
BTA41-1200BW	Q1	TO-3P-3_L15.3-W4.5-P5.52-L	1	BTA41-1200BW	KTP
10k Ω	R1	R0805	1	0805W8F1002T5E	UNI-ROYAL
100 Ω	R2	RES-SMD_L6.4-W3.2-R2512	1	SR2512FK-7W100RL	YAGEO
Potentiometer 250k Ω -500k Ω	SETPOINT	WH160-1-104	1	Potentiometer WH160-1-104	JML
Heat sink 200-250 $\times$ 120-150 $\times$ 70-90 $\leq 0.5^{\circ}\text{C/W}$	U1	HEATSINK-TH_XSD366-094	1	XSD366-094	XSD

(40A, 1200V) has been selected to support operation under high-load conditions.

A 250k Ω -500k Ω potentiometer is provided for output voltage control. The MOC3041 optocoupler implements the solid-state relay function and provides optical isolation between the low-voltage control section and the high-voltage AC mains, improving safety and noise immunity.

Voltage control

The circuit controls the AC output voltage using the phase-angle control technique. The potentiometer, together with the 0.1 μF capacitor, forms an RC timing network that

determines the charging rate of the capacitor during each half-cycle of the AC input.

When the capacitor voltage reaches the DIAC breakdown voltage, approximately 32V, the DIAC conducts and delivers a sharp trigger pulse to the BTA41 TRIAC gate, turning it on. By varying the potentiometer setting, the TRIAC's firing angle is adjusted within each AC half-cycle. Fig. 4 shows the integration of the potentiometer into the circuit.

If the TRIAC is triggered late in the half-cycle, only a small portion of the sine wave is applied to the load, resulting in a lower RMS voltage and reduced power. Conversely, if the TRIAC is triggered early, a larger

portion of the sine wave reaches the load, producing a higher RMS voltage. In this manner, smooth and continuous control of the AC output voltage, from near zero to full mains voltage, is achieved by adjusting the potentiometer.

The operating range is based on the specifications provided in the manufacturer's datasheet.

The assembled circuit described here was tested using a household power line with a maximum voltage of 260V and a load of up to 30A. Higher voltage and current ranges were not tested. The stated ratings therefore rely on the claims made in the TRIAC datasheet.

How complete circuit works?

For clarity, the circuit operation is divided into two modes: dimmer/voltage control mode and solid-state relay mode. Mode selection is determined by jumpers JP1 and JP2.

Working in dimmer/voltage control mode

When solder jumper JP1 is configured for dimmer (voltage control)